TEAM REGULAR
CMPS 4700 PA1 Report
Alameen Adeku, Sirjana Acharya, Harper Laskie.

Team Members & Roles

Team Member	Role	Status
Harper Laskie	Worked on preprocessing data and generating plots.	Completed
Sirjana Acharya	Worked on generating data before & after preprocessing	Completed
Alameen Adeku	Assembled code and worked on the report.	Completed

PRE PROCESSING PIPELINE

Step 1 : Data Loading
The raw CSV was loaded and inspected for shape and missing values like so:

```
df = pd.read_csv('smmh.csv')
print(f"Raw dataset shape: {df.shape}")
print(f"Missing values:\n{df.isnull().sum()}")
```

Step 2: Column Renaming
All 20 survey question columns were renamed from verbose question strings to concise identifiers (Age, Gender, Q9–Q20, etc.) to simplify downstream processing. Unused columns — Timestamp, platform usage binary, and platform selection — were dropped.

Step 3 — Missing Value Handling
The Organization column contained approximately 30 missing entries, which were imputed with the label 'Unknown' to preserve row count and avoid information loss like so:

```
df['Organization'] = df['Organization'].fillna('Unknown')
```

Step 4 — Gender Normalization
The Gender column contained inconsistent free-text entries (e.g., 'male', 'M', 'MALE'). A normalization function was applied to standardize all values into three clean categories — Male, Female, and Other.

Step 5 — Daily Usage Encoding
The Daily_Usage column contained ordinal text ranges ('Less than an Hour', 'Between 1 and 2 hours', etc.). These were mapped to integers 0–5 to preserve the ordinal relationship while making the column numeric:

Original Label	**Encoded Value**
Less than an Hour	0
Between 1 and 2 hours	1
Between 2 and 3 hours	2
Between 3 and 4 hours	3
Between 4 and 5 hours	4
More than 5 hours	5

Step 6 — Target Variable Construction
As detailed in Section 2, the MH_Score composite was calculated from Q9–Q20, binned into three risk tiers, and stored as the TARGET column. The raw MH_Score column was then dropped to prevent leakage.

Step 7 — One-Hot Encoding
Four categorical columns — Gender, Relationship_Status, Occupation, and Organization — were one-hot encoded using pd.get_dummies. This expanded the feature space from 19 columns to 45 columns, as each unique category value becomes its own binary indicator column.

Step 8 — Label Encoding the Target
The TARGET column text labels were converted to integers using LabelEncoder. The alphabetical mapping is:

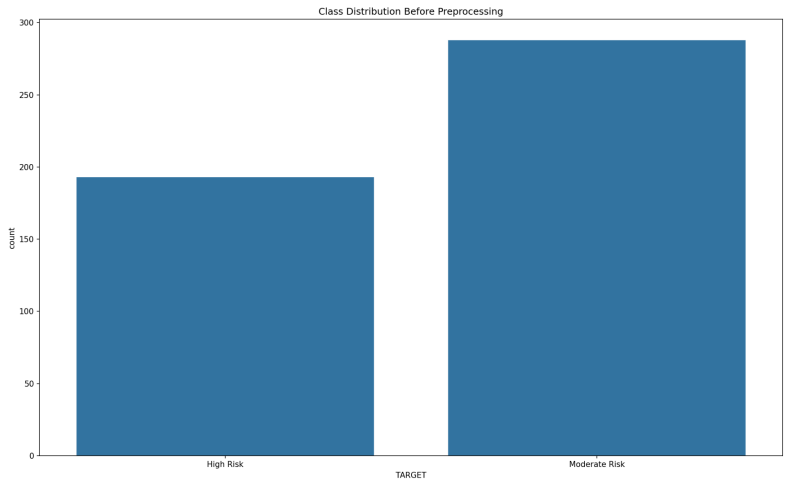
High RIsk - 0
Low Risk - 1
Moderate RIsk - 2

Step 9 — MinMax Normalization
All continuous and ordinal numeric columns — Q9 through Q20, Age, and Daily_Usage — were scaled to the [0, 1] range using MinMaxScaler. This is required for distance-sensitive models (K-NN, SVM) to prevent columns with larger numeric ranges from dominating similarity calculations.
One-hot encoded columns were excluded as they are already binary (0/1). The TARGET column was never scaled.

Step 10 — Output Reformatting
The dataframe was restructured into a standardized template with SAMPLE_ID as the first column, TARGET as the second, and all feature columns renamed generically to A1, A2, ..., An.

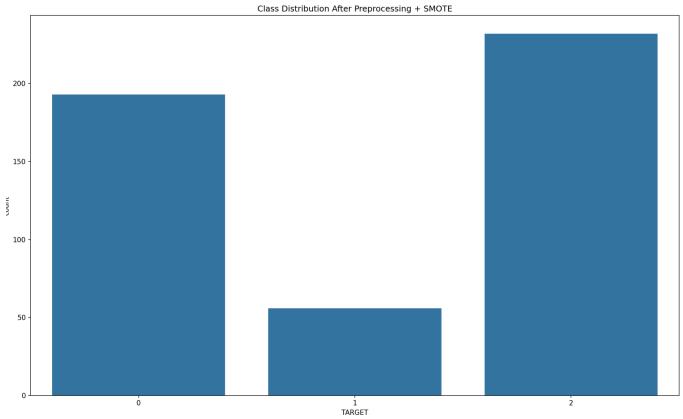
VISUALIZATIONS AND PLOTS

1. **Class Distribution Before Preprocessing Plot**
    a. No participant has a score <= 25
    b. There no “Low Risk” because the targets were not all above our determined ‘low risk’
    c. SMOTE creates synthetic examples for all classes, so now it exists, giving us a broader range than just the ‘moderate/high’ while keeping it within our ranges.



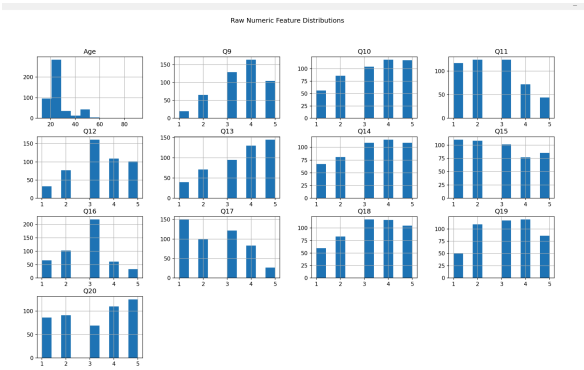
2. Class Distribution After Preprocessing + SMOTE

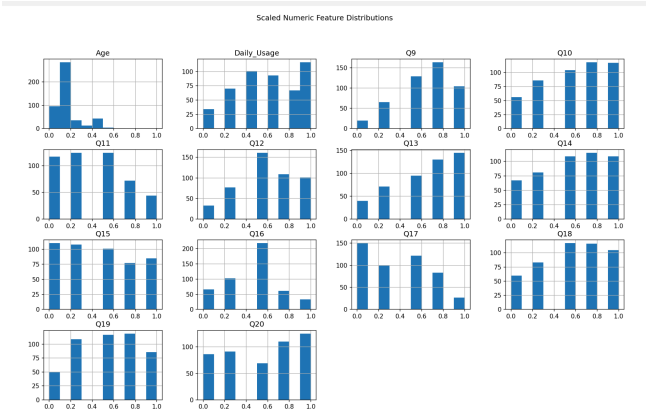
a. 0 = High
b. 1 = Low
c. 3 = Moderate

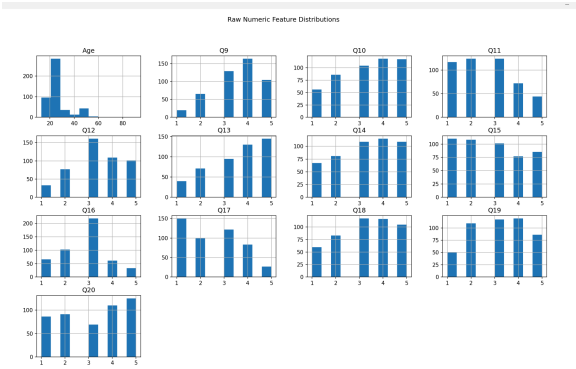


3. Raw Numeric Feature Distributions:

a. x-axis : the actual values of each feature in the raw data
   i. Age might go 13-91, Daily_Usage mapped 0-5, thus Q9-Q20 scores: 1-5
b. Y-axis: the count of participants falling in each section of the x-axis
   i. If Age 20-30 has y=15, then 15 participants are aged 20-30

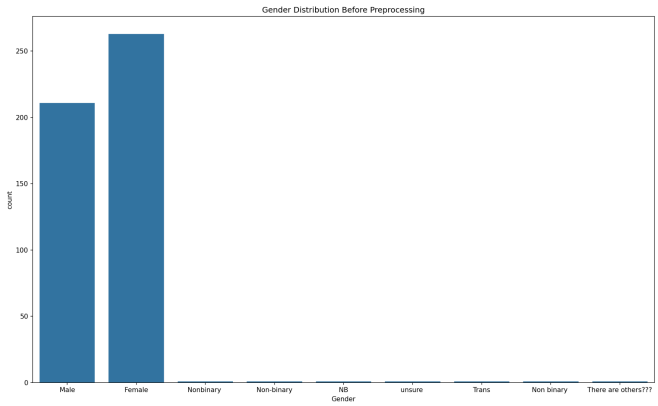
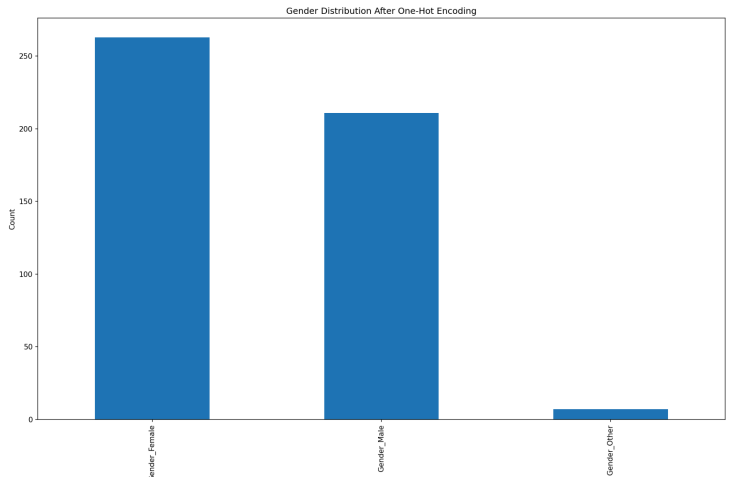






4. Scaled Numeric Feature Distributions:

a. Applied MinMaxScaler for 0-1 ranges
b. X-axis shows values scaled from 0-1
   i. 0 = minimum value in original column
   ii. 1 = max value in original column
c. y-axis : count of participants in each bin

5. One Hot Encoding

Applied one hot encoding to Gender, Relationship_Status, Occupation, Organization to enable categorical text columns into numbers so your ML models can work with them. The histogram shows the distribution for gender before and after one hot encoding was applied.